Practice workbook · Area of a triangle using half ab sin C

Using the area formula (1/2)ab sin C for a triangle, forwards or backwards, including the case where a given area allows an acute or an obtuse included angle.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-11

1. Question 1 · part whole · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part whole only.

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1 In triangle ABC, $AB = 2x$ cm, $BC = 3x$ cm and $AC = 4x$ cm

The area of triangle ABC is 50 cm^2

Find, to 2 decimal places, the value of x

(4)

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(Total for Question 1 is 4 marks)

1. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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2. Question 2 · part b · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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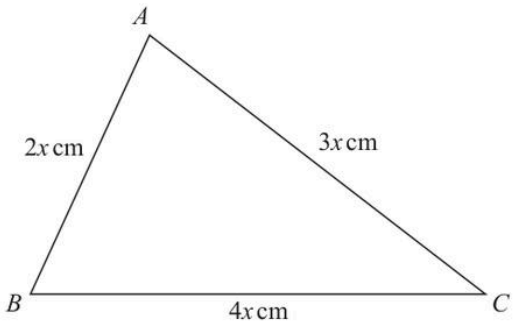


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle ABC in which

$AB = 2x$ cm $\quad AC = 3x$ cm $\quad BC = 4x$ cm

(a) Show that $\sin ABC = \frac{3\sqrt{15}}{16}$

(4)

Given that the area of triangle ABC is $\frac{75\sqrt{15}}{64}$ cm^2

(b) find the value of x.

(2)

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2. Question 2 · part b · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 2 continued

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(Total for Question 2 is 6 marks)

2. Question 2 · part b · 2 marks · Working and answer

Working and answer

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3. Question 2 · part b · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

2 In triangle ABC, $AB = 3x$ cm, $BC = 5x$ cm and $\angle ABC = 110°$

(a) Find, in degrees to one decimal place, the size of $\angle BCA$

(4)

The area of triangle ABC is 24 cm^2

(b) Find, to 3 significant figures, the value of x

(3)

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3. Question 2 · part b · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

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Question 2 continued

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(Total for Question 2 is 7 marks)

3. Question 2 · part b · 3 marks · Working and answer

Working and answer

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4. Question 3 · part b · 4 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

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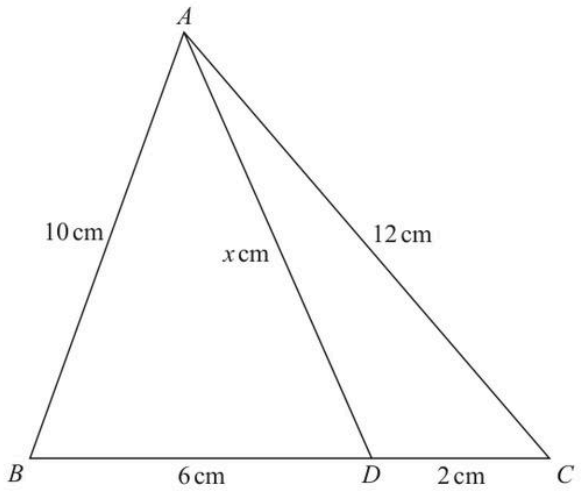


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the triangle ABC in which $AB = 10\,\text{cm}$ and $AC = 12\,\text{cm}$.
The point D lies on BC such that $BD = 6\,\text{cm}$, $DC = 2\,\text{cm}$ and $AD = x\,\text{cm}$.

(a) Show that $x = 11$

(4)

(b) Find the area, in cm^2 to 3 significant figures, of triangle ADB.

(4)

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4. Question 3 · part b · 4 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 8 marks)

4. Question 3 · part b · 4 marks · Working and answer

Working and answer

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5. Question 3 · part a · 3 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

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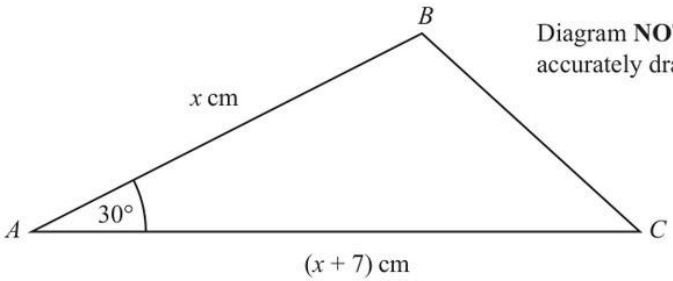


Figure 1

Figure 1 shows triangle ABC where

$AB = x$ cm $\quad AC = (x + 7)$ cm $\quad \angle BAC = 30°$

The area of triangle $ABC = 36$ cm^2

(a) Show that $x = 9$ **(3)**

(b) Find, in cm to 3 significant figures, the length of BC **(2)**

(c) Find, in degrees to one decimal place, the size of

(i) $\angle ABC$

(ii) $\angle ACB$ **(3)**

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5. Question 3 · part a · 3 marks · Working and answer

Working and answer

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6. Question 5 · part b · 3 marks

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Further practice · Attempt part b only.

5 In triangle ABC, $AC = 12$ cm, $BC = 14$ cm, $AB = x$ cm and angle $ABC = 30°$

(a) Show that $x = P\sqrt{3} \pm \sqrt{95}$ where P is a prime number.

(5)

(b) Hence or otherwise, find in cm^2, the difference between the two possible areas of triangle ABC

Give your answer in the form $m\sqrt{n}$ where m is a prime number and n is an integer.

(3)

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6. Question 5 · part b · 3 marks

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Further practice · Attempt part b only.

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Question 5 continued

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(Total for Question 5 is 8 marks)

6. Question 5 · part b · 3 marks · Working and answer

Working and answer

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7. Question 6 · part c · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

6 Given that $\dfrac{a+\sqrt{5}}{\sqrt{5}-2} = 11 + 5\sqrt{5}$

(a) without using a calculator, find the value of a
Show your working clearly.

(2)

Triangle PQR is such that

$PR = (x+3)\,\text{cm}$ $QR = x\,\text{cm}$ angle $QPR = 30°$ angle $PQR = 45°$

(b) Show that $x = 3 + 3\sqrt{2}$

(3)

Given that $\sin 105° = \dfrac{\sqrt{6}+\sqrt{2}}{4}$ and that the area of triangle PQR is $A\,\text{cm}^2$

(c) find the exact value of A in the form $\dfrac{9}{8}\left(p\sqrt{6} + q\sqrt{2} + r\sqrt{3} + s\right)$

where p, q, r and s are integers.

(3)

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7. Question 6 · part c · 3 marks

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Further practice · Attempt part c only.

Question 6 continued

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7. Question 6 · part c · 3 marks

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Further practice · Attempt part c only.

Question 6 continued

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7. Question 6 · part c · 3 marks

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Further practice · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 8 marks)

7. Question 6 · part c · 3 marks · Working and answer

Working and answer

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8. Question 3 · part c · 4 marks

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Further practice · Attempt part c only.

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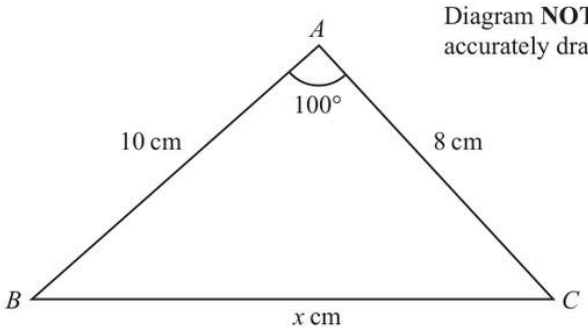


Figure 2

Figure 2 shows triangle ABC where

$$AB = 10 \text{ cm}, \quad AC = 8 \text{ cm}, \quad BC = x \text{ cm} \quad \text{and} \quad \angle BAC = 100^\circ$$

(a) Find, to 3 significant figures, the value of x

(2)

(b) Find, in degrees to one decimal place, the size of

(i) angle ABC (ii) angle ACB

(3)

The bisector of angle ABC meets AC at the point M

(c) Find the area, in cm^2 to 3 significant figures, of triangle BMC.

(4)

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8. Question 3 · part c · 4 marks

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Further practice · Attempt part c only.

Question 3 continued

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(Total for Question 3 is 9 marks)

8. Question 3 · part c · 4 marks · Working and answer

Working and answer

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9. Question 8 · part b · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part b only.

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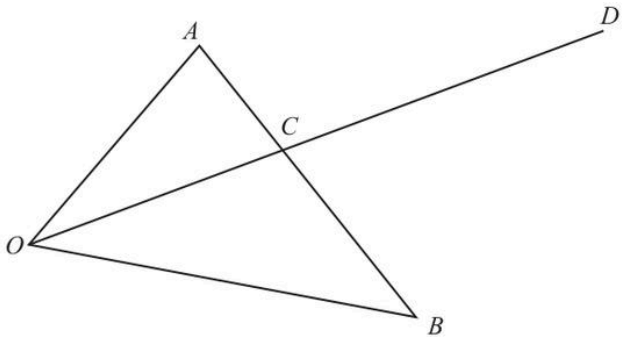


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle AOB

$\overrightarrow{OA} = 4\mathbf{a} + 5\mathbf{b}$ $\overrightarrow{OB} = 8\mathbf{a} - \mathbf{b}$ $\overrightarrow{OD} = 15\mathbf{a} + 10\mathbf{b}$ where $|\mathbf{a}| = |\mathbf{b}| = 1$

(a) (i) Find $\overrightarrow{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

(ii) Find, in its simplest form, the exact value of $|\overrightarrow{AB}|$

(3)

(b) Find the area of triangle AOB

(4)

The point C lies on AB and OD such that O, C and D are collinear.

(c) Use a vector method to find vector $\overrightarrow{OC}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$

(5)

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9. Question 8 · part b · 4 marks · Working and answer

Working and answer

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10. Question 3 · part b · 3 marks

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Further practice · Attempt part b only.

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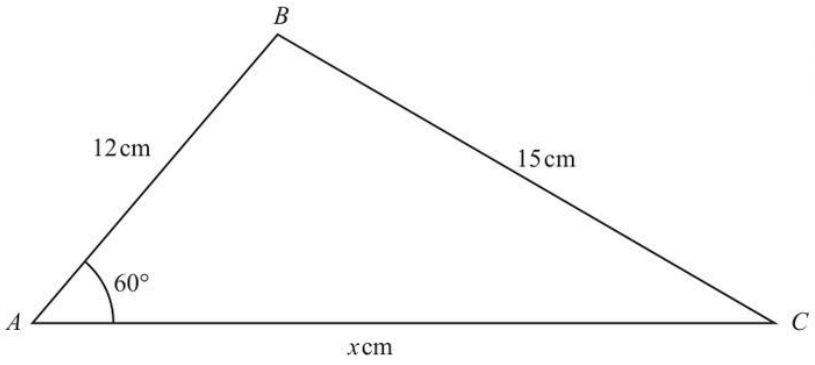


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Figure 1

Figure 1 shows triangle ABC where

$AB = 12$ cm $\quad BC = 15$ cm $\quad AC = x$ cm $\quad \angle BAC = 60°$

(a) Show that x satisfies the equation

$$x^2 - 12x - 81 = 0$$

(3)

(b) Hence find the exact area, in cm^2, of triangle ABC

Give your answer in the form $9\left(p\sqrt{q} + \sqrt{r}\right)$ where p, q and r are integers.

(3)

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10. Question 3 · part b · 3 marks

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Further practice · Attempt part b only.

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Question 3 continued

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(Total for Question 3 is 6 marks)

10. Question 3 · part b · 3 marks · Working and answer

Working and answer

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11. Question 3 · part whole · 5 marks

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Further practice · Attempt part whole only.

3 In triangle ABC, $AB = 11\,\text{cm}$ and $BC = 12\,\text{cm}$.

The area of triangle $ABC = 33\,\text{cm}^2$

Find, in cm to 3 significant figures, the two possible lengths of AC.

(5)

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11. Question 3 · part whole · 5 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 5 marks)

11. Question 3 · part whole · 5 marks · Working and answer

Working and answer

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12. Question 7 · part b · 7 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

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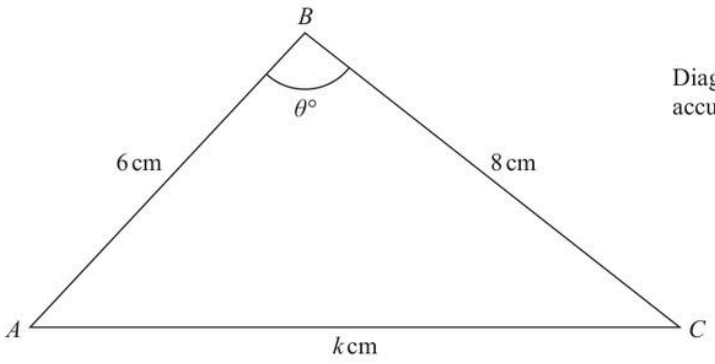


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Figure 2

Figure 2 shows triangle ABC

$AB = 6\,\text{cm}$ $BC = 8\,\text{cm}$ $AC = k\,\text{cm}$ $\angle ABC = \theta°$

(a) Show that $\cos\theta° = \dfrac{100 - k^2}{96}$ (2)

The area of triangle ABC is $\sqrt{455}$ cm^2

(b) Find the two possible values of k (7)

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12. Question 7 · part b · 7 marks

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Further practice · Attempt part b only.

Question 7 continued

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12. Question 7 · part b · 7 marks

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Further practice · Attempt part b only.

Question 7 continued

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12. Question 7 · part b · 7 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

Question 7 continued

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(Total for Question 7 is 9 marks)

12. Question 7 · part b · 7 marks · Working and answer

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